



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

Part A: Introduction										
1	Course Title	Data Structure and Algorithms – I Laboratory								
2	Course Code	CSE 2216								
3	Pre-requisites	CSE 1112, CSE 1116								
4	Course Type	Core Course								
5	Credit Hours	1.00								
6	Contact Hours	2.5 Hours/Week								
7	Semester	4 th								
8	Total Marks	100								
9	Course Instructor’s Information	Adiba Shaira Email: adiba@cse.uiu.ac.bd Room: 419								
10	Course Rationale	This course has been designed to provide a solid foundation about the data structure and algorithms used in computer science. This course will give insights about the pros and cons of different data structures and algorithms.								
11	Course Objectives	The objectives of this course are: • To familiarize the basic data structures (array, linked list). • To familiarize complex data structures (queue, stack, priority queue) using basic data structures • To use suitable data structures for different algorithms • To introduce the algorithms and their complexity and use cases								
Part B: Content of the Course										
12	Course Contents (approved by UGC)	Sorting Algorithms: Bubble Sort, Insertion Sort, Selection Sort, Linked List: Single Linked List, Double Linked List, Stack: Implementation using Array and Linked List, Queue: Implementation using Array and Linked List, Binary Search Tree: Construction, Operations (Insertion, Deletion, etc.), Graph: Introduction, Implementation using Adjacency Matrix and Adjacency List, BFS, DFS. Tree Traversal(Preorder, Postorder, Inorder)								
13	Course Outcomes (COs)	<table><tr><th>COs</th><th>Description</th></tr><tr><td>CO1</td><td>Implement appropriate data structure to handle large datasets efficiently as applied to specified problem definition.</td></tr><tr><td>CO2</td><td>Able to handle operations like searching, insertion, deletion, traversing mechanism etc. on various data structures.</td></tr><tr><td>CO3</td><td>Able to use linear and non-linear data structures like stacks, queues, linked list etc.</td></tr></table>	COs	Description	CO1	Implement appropriate data structure to handle large datasets efficiently as applied to specified problem definition.	CO2	Able to handle operations like searching, insertion, deletion, traversing mechanism etc. on various data structures.	CO3	Able to use linear and non-linear data structures like stacks, queues, linked list etc.
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14 Mapping of COs and Program outcomes										

COs	Program Outcomes(POs)											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C01			C									
C02			C									
C03				C								

15	Mapping COs with Teaching-Learning and Assessment Strategy											
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Class	Topics/Assignments	Course Outcomes (COs)	Lab Outcomes/ Activities
1	Introduction Sorting Algorithms : Bubble Sort	C01	Lecture
2	Sorting Algorithms: Selection Sort, Insertion Sort	C01	Lecture
3	Coding Test 1: Sorting Searching: Linear. Binary	C01, C02	Coding Test, Lecture
4	Coding Test 2: Searching Singly Linked List: Intro + Insertion + Deletion + Search	C02, C03	Coding Test, Lecture
5	Doubly Linked List, Circular Linked List, <i>Declaration of Home Assignment 1: Linked List</i>	C02, C03	Lecture
6	Mid-Term Exam	C01,C02, C03	Graded Exam
7	Stack using Array and Linked List, Home Assignment 1: Linked List Evaluation	C02, C03	Lecture, Home Assignment
8	Queue using Array and Linked List, <i>Declaration of Home Assignment 2: Stack Queue</i>	C02, C03	Lecture
9	Graph Representation, Home Assignment 2: Stack Queue Evaluation	C01,C02, C03	Lecture, Home Assignment
10	Coding Test 3: Graph Representation BFS, DFS, <i>Declaration of Home Assignment 3: BFS DFS</i>	C02, C03	Coding Test, Lecture
11	Tree Traversal(Preorder, Postorder, Inorder), BST, Home Assignment 3: BFS DFS Evaluation	C01,C02, C03	Lecture, Home Assignment
12	Final Exam	C01, C02, C03	Graded Exam

Part C: Assessment and Evaluation Methods

Assessment Types	Marks
Attendance	10%
Home Assignments	25%
Coding Tests	30%
Mid Exam	15%
Final Exam	20%

Grading System

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Part D: Learning Resources

Text Book	1. Introduction to Algorithms – Thomas H. Cormen (4 th edition, MIT Press & McGraw Hill, 2022) 2. Data Structure and Algorithms in C++ - Goodrich, Tamassia (2 nd edition, John Wiley and Sons Inc., 2003)
Reference	1. http://www.geeksforgeeks.org (for implementation)
LMS URL	http://lms.uiu.ac.bd/course/view.php?id=2459

Appendix-1: Program outcomes

POs	Program Outcomes
PO1	An ability to apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PO3	An ability to design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PO4	An ability to conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PO5	An ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations
PO6	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO8	An ability to apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PO9	An ability to function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.

PO10	An ability to communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PO12	An ability to recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.